No Rule Can Be Fixed

Floating Point as the Compiled Solution to Continuation Applied to Rules

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Every quantity is quoted verbatim from execution. The central empirical anchor is a standard that has been in industrial use since 1985 and was arrived at from the opposite direction, by four decades of numerical failure; that convergence is offered as the paper's principal evidence. Corrections made during preparation are logged in place, and the companion audit paper records the program's full retraction ledger.

Abstract

This paper applies a continuation constraint to rules themselves and derives a consequence that turns out to be already implemented in every computer manufactured since the mid-1980s.

The constraint is minimal. Distinction exists (C0), and every admissible state must have a successor distinct from itself (C1). Fixed points are excluded. The paper first establishes the base case exhaustively: on two states there is exactly one C1-legal map and it is the bijection, so reversibility is not an axiom but a consequence at the floor, and the gap between legal and reversible opens only above two states — 8 legal against 2 bijective at three states, 81 against 9 at four.

The extension is then made. A rule is itself a state in the space of rules, so C1 applies to it. A rule that cannot move is a fixed point and is forbidden. Rules must therefore relocate, and relocation is not the failure of a rule but the mechanism by which it remains legal. The consequence is that no system containing its own rules can be complete, because completion would pin the rule that permits continuation.

The empirical anchor is the binary floating-point format. A fixed-point format dies at both boundaries: with sixteen bits in a Q8.8 layout, values above 127.996 cannot be represented at all and values below 0.0039 flush to zero, so continuation fails at the top and distinction fails at the bottom. Letting the radix point move buys a dynamic range 3.275e+04 times larger at

identical bit count. The format carries two independent registers — a significand and an exponent — and their separation is exact under radix scaling, holding a significand at 0.750000 across scalings by two hundred powers of two while the exponent alone varies. Renormalization, the act of moving the point, occurs on 38.7% of random multiplications: the main path, not an error path. And where the point can no longer move, in the denormal region, precision freezes at 4.941e-324 and the value flushes to zero. The system dies exactly where its rule became fixed.

A structural bridge is then established between this and the integers. Writing an integer in its gnomon coordinate as a shell index plus an offset gives a representation in which the offset is bounded by twice the shell and the shell is unbounded — a significand and an exponent. Verified over 199,999 consecutive integers, the offset never leaves its bound. In value coordinates the step between shells grows without limit as 2k+1; in shell coordinates the step is always exactly one. The apparent jumps of quadratic growth are a projection artifact of the value coordinate, and the integers admit a normalized representation in which growth is smooth.

The paper closes on the balance condition. A rule may be bent for as long as balance is maintained; normalization is exactly that permitted bending, and denormalization is what happens when the balance is lost. Incompleteness is therefore not a defect of the systems that have it. It is the license under which they continue to exist.

1 Introduction

The question this paper addresses is what happens when a continuation requirement is applied not to the states of a system but to the rules that govern it.

The requirement is the one that structures the wider program: every admissible state must have a successor distinct from itself. Applied to states, it forbids dead ends and produces a long chain of consequences. Applied to rules, it produces something initially uncomfortable — that a rule which cannot change is inadmissible on the same grounds as a state which cannot change. This appears to dissolve the notion of a rule entirely, since a rule that changes seems to be no rule at all.

It does not dissolve it. There is a third option between a rule that is pinned and a rule that is abandoned, and the third option is a rule that relocates while remaining in force. The paper's claim is that this option is not merely available but mandatory, and that it has a precise and familiar implementation.

The implementation is binary floating-point arithmetic. The standard governing it was ratified in 1985 after roughly two decades of accumulating numerical failures in scientific computing — overflow disasters, catastrophic cancellation, machine-dependent results — and it was arrived at empirically, one repaired failure at a time. Its central design decision is that the radix point does not occupy a fixed position and is relocated on every operation that requires it. The paper argues that this is the same structure the continuation constraint forces, reached from the opposite direction, and treats the convergence as evidence: when a constraint argument and four decades of engineering pain arrive at an identical structure, the constraint reading has support it did not manufacture for itself.

Structure. Sections 2 and 3 establish the primitives and the base case. Section 4 makes the extension to rules. Sections 5 through 8 develop the floating-point result. Section 9 establishes the bridge to the integers. Sections 10 and 11 state the incompleteness consequence and the balance condition. Section 12 records what remains open. All measurements are reproduced in the appendix.

2 The Primitives

C0 (Distinction). At least two states exist that are not identical. This is the minimum condition for a system to carry any structure at all; with a single state there is nothing to distinguish and no information can exist.

C1 (Continuation). Every admissible state admits at least one successor distinct from itself. A self-loop is not a successor for this purpose. Terminal states are excluded.

One clarification is needed immediately, because the rest of the paper depends on it. C1 does not forbid a value from being held constant. A system with more than one channel may freeze one channel while continuing in another, and such a state has a successor distinct from itself in the full state space. What C1 forbids is a state frozen in every channel simultaneously, which has no successor at all. A committed measurement is legal; a completely committed system is not.

This distinction is what makes rules interesting. A rule that holds a value fixed is unproblematic. A rule that is itself unable to move is a different object, and Section 4 shows it falls under the prohibition.

3 The Base Case: Reversibility Is Free at the Floor

Before extending C1 to rules, its behaviour on the smallest possible system is established exhaustively, because the extension's force depends on how tightly C1 binds when there is nowhere to hide.

Consider all maps from a finite state set to itself, and retain those with no fixed point — those satisfying C1 at every state. Counting the survivors, and counting how many of the survivors are bijections:

States	C1-legal maps	Bijective among them	Gap
2	1	1	**0**
3	8	2	6
4	81	9	72

Theorem (base case). *On two states there is exactly one C1-legal map, and it is the bijection.*

Proof. There are four maps on two states. Three have a fixed point: the identity fixes both, and each of the two constant maps fixes its own image. The remaining map exchanges them. It has no fixed point and it is a bijection. Exhaustive over the four cases. Square.

Two consequences follow, and both matter later.

Reversibility is not an axiom at the floor. It does not need to be assumed; C1 alone forces it when there are only two states. The gap between "legal" and "reversible" is zero at the base and opens only above it — six maps wide at three states, seventy-two at four. A program that introduces reversibility as an independent primitive has added an axiom it did not need, and the wider program has withdrawn that axiom accordingly.

Direction is not in the step. Applying the exchange twice returns the original state, so the state two steps ahead and the state two steps behind coincide. A single step carries no arrow. An arrow appears only once steps accumulate into an asymmetry, which means temporal direction is a property of the accumulated history and not of the transition. This will matter in Section 10, where completion is shown to be the destruction of exactly that accumulation.

4 The Extension: C1 Applied to Rules

A rule is a constraint on transitions. It is also, viewed one level up, a state — an element of the space of possible rules, occupied by a system at a given moment. Nothing in C1 restricts it to a particular level of description; it is a statement about admissible states in a transition system, and the space of rules is a transition system whose states are rules and whose transitions are revisions.

Proposition. *A rule that cannot be relocated is a fixed point in the space of rules and is therefore inadmissible under C1.*

The argument is direct. If a rule admits no successor distinct from itself, it is a fixed point by definition. C1 excludes fixed points. Therefore a rule with no admissible relocation is excluded.

The natural objection is that this appears to abolish rules altogether: if every rule must change, nothing constrains anything, and the framework collapses into permissiveness. The objection fails because it conflates two different things — a rule changing its content, and a rule changing its position. The third option identified in the introduction is precise:

$$\textit{pinned} \rightarrow \textit{forbidden} \qquad \textit{abandoned} \rightarrow \textit{no constraint} \qquad \textit{relocated} \rightarrow \textit{admissible}$$

A relocated rule is still in force. What has moved is where it applies, not whether it applies. The rest of this paper is an examination of a system in which this third option is implemented exactly, and in which the two failure modes at either end of it are directly observable.

5 A Pinned Rule Dies at Both of Its Own Boundaries

Fixed-point arithmetic pins the radix point at a fixed bit position. Take a signed sixteen-bit format with eight integer bits and eight fractional bits. The rule "the point sits between bit 7 and bit 8" cannot move.

Property	**Value**	**Failure**
Largest representable	127.99609375	128.0 cannot be represented at all
Smallest representable	0.00390625	0.001 flushes to zero
Dynamic range	**3.277e+04**	—

The two failures are not the same failure, and the difference is exactly the difference between the two primitives.

At the top, C1 fails. A value above the maximum has no representation. The system cannot continue upward; there is no successor state available. This is a dead end in the literal sense the constraint forbids.

At the bottom, C0 fails. A value below the minimum does not merely lose accuracy — it becomes zero, and becomes indistinguishable from every other value that also became zero. Distinction is destroyed. Two states that differed no longer differ.

A pinned rule therefore kills the system it governs at both of its own edges, and it kills it in the two distinct ways the two primitives forbid. The rule does not fail because the format is too small; the dynamic range is a consequence of the pinning, not of the bit count, as the next section demonstrates at identical width.

6 The Floating Solution, at Identical Cost

The alternative allocates the same sixteen bits differently: some to a significand and some to an exponent, with the radix point positioned by the exponent rather than by the format.

Format (16 bits)	Dynamic range	Ratio
Fixed point, Q8.8	3.277e+04	—
Binary floating point	1.073e+09	**3.275e+04 times larger**

The bit count is identical. The entire difference is whether the rule locating the radix point is permitted to move. Letting it move purchases four to five orders of magnitude of range at zero storage cost, which is the quantitative form of the claim that relocation is not a concession but a mechanism.

It is worth stating what has not happened here. No information has been created. The format still holds 2^16 distinct values; what changed is their distribution across the number line — dense near zero, sparse far from it. The relocation redistributed resolution rather than adding it. This is the conservation signature the wider program associates with relocation generally: the address moves, the content does not increase.

7 Two Independent Channels in One Word

The format carries two fields that vary independently, and their independence is measurable directly.

Holding the significand and varying the exponent. Scaling a value by powers of the radix leaves the significand bit-identical while the exponent alone changes:

Value	Significand	Exponent
0.75×2^{-200}	**0.750000**	−200
0.75×2^{-100}	**0.750000**	−100
0.75×2^{0}	**0.750000**	0
0.75×2^{100}	**0.750000**	+100
0.75×2^{300}	**0.750000**	+300

Holding the exponent and varying the significand. Sweeping the significand from 0.50 to 0.99 at a pinned exponent of 11 produces values from 1024.000 to 2027.520 with the exponent spread across the sweep measured at exactly 0.

The significand is a value channel: it carries what the quantity is within its scale. The exponent is a shape channel: it carries the scale itself. Each moves with the other pinned. Two registers, one word.

Correction, logged in place. An earlier statement of this result claimed the significand holds constant across a hundred orders of magnitude, which implies decimal scaling. It does not. Measured under scaling by powers of ten, the significand drifts from 0.508 to 0.951 — the two channels mix. The separation is exact under radix scaling and only under radix scaling. The corrected claim is narrower and is the one used here: scaling by the radix moves the exponent field alone. This correction was made during the preparation of this paper and is recorded in the companion audit ledger.

8 Normalization: The Rule Breaking Itself to Remain Legal

If relocation were rare — an exceptional repair invoked at the boundaries — the argument of Section 4 would be weak, because the rule would be effectively fixed across the operating range and merely patched at the edges. It is not rare.

Measurement	Result
Random multiplications tested	100,000
Requiring exponent renormalization	**38,691**

Measurement	Result
Fraction of operations	**38.7%**

On roughly two operations in five, the product of two normalized values is not itself normalized, and the point must move for the result to be representable in the format at all. Relocation is on the main path. It is what the arithmetic unit does as part of ordinary work, not what it does when something has gone wrong.

This is the precise sense in which a rule breaks itself in order to remain in force. The constraint "the significand lies within its normalized range" would be violated by the raw product; the system restores it by moving the point and adjusting the exponent. The rule is never suspended. Its position is updated so that it continues to hold. Breaking, here, is the mechanism of persistence rather than its failure.

A note on the earlier overstatement. This figure was previously reported as approximately 100% of operations. The measured value is 38.7%, independently reproduced here against an earlier run at 38.8%. The corrected figure supports the argument better than the overstatement did: a rule that relocated on every operation would be indistinguishable from having no rule, whereas one that relocates on two-fifths is visibly a constraint that is visibly being maintained.

9 Where the Point Can No Longer Move

The prediction of Section 4 is that a system dies where its rule becomes fixed. The format provides a region in which exactly this occurs and can be measured, because below a certain magnitude the exponent has no further room to decrease and the point stops floating.

Value	Normalized	Precision (unit in last place)
1.000e−300	yes	1.658e−316
1.000e−310	no	**4.941e−324**
1.000e−320	no	**4.941e−324**
4.941e−324	no	**4.941e−324**
0.000e+00	—	**distinction gone**

Above the normal threshold the precision scales with the magnitude: the point floats, and the gap between representable neighbours shrinks as values shrink. Below it, the precision freezes at a single absolute value and stops improving no matter how small the number becomes. The exponent has bottomed out; the only remaining way to represent smaller values is to spend significand bits, and when those are exhausted the value flushes to zero and becomes indistinguishable from every other value that also flushed.

The death is by pinning, not by wear. Nothing degrades in the denormal region. No noise accumulates, no component fatigues. The single thing that changes is that the rule locating the point loses its freedom to move, and the system fails immediately afterward — first losing resolution, then losing distinction entirely. The two failures of Section 5 reappear here in miniature: continuation degrades, then distinction is destroyed. The prediction and the measurement agree.

10 The Bridge: Integers in Normalized Form

The preceding sections concern a designed artifact. This section shows the same structure in an object nobody designed.

Any positive integer can be written in gnomon coordinates as a shell index and an offset within that shell, where the shell is the integer part of the square root and the offset is the remainder above the largest square below:

$$n = k^2 + m, \quad k = \lfloor \sqrt{n} \rfloor, \quad 0 \le m \le 2k$$

The offset is bounded by twice the shell; the shell is unbounded. This is the structure of a floating-point representation. The offset is a significand — a bounded position within a scale. The shell is an exponent — an unbounded selector of the scale itself.

The bound is exact. Tested over every integer from 1 to 199,999, the offset left its permitted range in zero cases. The significand never overflows its bound, and the carry into the next shell lands exactly.

The consequence is a change in what growth looks like depending on which coordinate reports it:

n	Shell k	Offset m	Step in value space	Step in shell space
8	2	4	5	**1**
24	4	8	9	**1**

n	Shell k	Offset m	Step in value space	Step in shell space
48	6	12	13	**1**
n	k	m	2k+1, unbounded	**always exactly 1**

In value coordinates the transition from one shell to the next costs 2k+1, and that cost grows without bound. This is the familiar quadratic jump, and read in that coordinate the sequence of squares looks discontinuous and accelerating. In shell coordinates the same transition costs exactly one, at every scale, forever.

The jumps are a projection artifact. Nothing in the integers is jumping. A coordinate was chosen in which the increment grows, and the growth was attributed to the object rather than to the coordinate. In the normalized coordinate the same structure advances by one step per shell with no acceleration anywhere. This is what a floating-point unit does to magnitude: it converts an unbounded range into a bounded significand plus a scale selector, and thereby makes uniform what was ragged.

The bridge is offered as a structural correspondence and not as an identity claim. What is established is that the gnomon decomposition has the significand-plus-exponent form, that its significand is provably bounded, and that its renormalization is exact. What is not established is that every property of the arithmetic format transfers.

11 Incompleteness as Mechanism

The results above support a single statement about completeness.

Proposition. *No system that contains its own rules can be complete, because completion would pin the rule that permits continuation.*

The argument assembles from what precedes it. A complete description of a system includes a description of its rules. If that description is exhaustive, then the rules have no remaining freedom to relocate — every position they might occupy is already specified. By Section 4, a rule with no admissible relocation is a fixed point in the rule space, and by C1 it is forbidden. A completed system is therefore an inadmissible one.

The denormal region of Section 9 is this proposition made concrete and measurable. In that region the exponent is fully determined — it has no remaining values to take. The rule locating the point is, locally, complete. And precisely there the system stops being able to distinguish and then stops being able to represent anything at all. Completion of the rule and death of the system are observed at the same threshold.

The consequence is a reversal of the usual reading of incompleteness. An incomplete system is normally understood as one that falls short of a description it ought to have. Under the constraint developed here, incompleteness is the condition under which a system is permitted to continue: the unspecified freedom in its rules is the room those rules need in order to move, and a system that used up that room would satisfy the definition of completeness by ceasing to satisfy the requirement of continuation.

Scope, stated plainly. This is a statement about systems governed by the continuation constraint adopted here, and it is not offered as a proof of, or a substitute for, the formal incompleteness theorems of mathematical logic. Those results concern the provability of sentences within formal systems of sufficient arithmetic strength and are established by entirely different means. Any relationship between the two is unexamined here and is listed in Section 12 as open. The proposition above stands or falls on C1, not on Gödel.

12 The Balance Condition

The permission to relocate is not unlimited, and the format makes the limit explicit.

Renormalization is legal for as long as the significand can be returned to its normalized range. That is the balance condition. Within it, the point may move as often as required — 38.7% of operations, as measured — and the rule remains in force throughout. Outside it, in the denormal region, the point cannot be returned to range, and the failures of Section 9 follow immediately.

bend freely, provided the significand stays in range → *normalization*

range exhausted → *denormalization* → *precision frozen* → *distinction lost*

The formulation this supports is that a rule may be bent for as long as balance is maintained. This is not a licence for arbitrary revision. It is a specific and measurable allowance: the rule's position may change without limit provided the quantity it governs remains within the bound that makes the rule meaningful. Normalization is that allowance exercised. Denormalization is what the same system looks like once the allowance is spent.

The same condition appears in the integer bridge. The offset may take any value it likes provided it stays within twice the shell; when it would exceed that bound the shell increments and the offset renormalizes to zero. Verified across 199,999 consecutive integers with zero violations, the bound is never breached — not because a mechanism enforces it externally, but because the coordinate is constructed so that exceeding it is the same event as advancing the scale.

13 Convergent Derivation

One methodological observation is worth recording, because it is the paper's strongest evidence and it is not of the paper's own making.

The floating-point standard was not derived from a constraint argument. It was assembled from failures: overflow in scientific codes, catastrophic cancellation, irreproducible results across machines, and the accumulated cost of numerical bugs across roughly two decades of practice. Its central decision — that the radix point moves — was reached because pinning it kept breaking things, one concrete disaster at a time.

The argument of Sections 2 through 4 reaches the same structure from a constraint: a rule that cannot move is a fixed point, fixed points are excluded, therefore rules must relocate. No numerical failures are consulted and no engineering history is used.

The two derivations agree on the structure, on the two-channel decomposition, and on the failure mode at the boundary. That agreement is the evidence. A constraint argument that reproduces the architecture of a standard arrived at independently by pressure — including the specific prediction that the system dies where its rule stops being able to move, which is the denormal region — has been tested against something it did not design and could not adjust. It is not proof. It is the kind of corroboration that a self-contained derivation cannot generate for itself, and the paper rests more weight on it than on any single measurement reported above.

14 Open Edges

Whether the relationship between this proposition and the formal incompleteness theorems is substantive or merely verbal. The two use the word for different objects and are established by different means. No claim of connection is made here and the question is untouched.

Whether the gnomon bridge extends beyond the structural correspondence. The significand-plus-exponent form, the bound, and the exact renormalization are established. Whether arithmetic on gnomon coordinates inherits the error behaviour of floating-point arithmetic is unexamined.

What sets the radix. In the format it is a design choice, and in the gnomon coordinate it emerges from the choice of squares. Whether any constraint selects a radix, or whether the choice is genuinely free, is not settled.

Whether the balance condition has a general form. It is stated here for two systems in which the bound is explicit. Whether an arbitrary constraint system admits a well-defined normalized range, and what determines its width, is open.

Whether the relocation quantum is universal. The wider program measures an exchange rate for one canonical relocation and finds it exact to machine epsilon across a range of widths; whether the floating-point renormalization obeys the same accounting has not been tested and would be a direct check of the two results against each other.

And, permanently, by the proposition itself: this paper cannot be complete. A finished account of the constraint would pin the constraint, and the argument given here forbids exactly that. The open edges above are not an admission of unfinished work. They are the condition under which the account remains admissible.

Status Ledger

Claim	Status	Basis (live measurement)
On two states, one C1-legal map and it is bijective	**PROVEN**	Exhaustive over 4 maps; 1 of 1
Legal–bijective gap opens above two states	**PROVEN**	8/2 at n=3, 81/9 at n=4
A pinned rule fails at both boundaries	**PROVEN**	Q8.8: overflow at 128.0, flush below 0.0039
Floating buys range at identical bit count	**PROVEN**	3.275e+04 times, 16 bits both
Two channels separate under radix scaling	**PROVEN**	0.750000 exact across 2^{-200} to 2^{300}
Channels separate under arbitrary scaling	**RETRACTED**	Base 10: drifts 0.508 to 0.951
Renormalization is on the main path	**PROVEN**	38,691 of 100,000 = 38.7%
Renormalization on ~100% of operations	**RETRACTED**	Measured 38.7%
The system dies where its rule is pinned	**PROVEN**	ulp freezes 4.941e-324, then zero

Claim	Status	Basis (live measurement)
Gnomon offset is a bounded significand	**PROVEN**	0 violations over n = 1..199,999
Shell-space growth is uniform	**PROVEN**	Step exactly 1 at every scale
C1 applies to rules as to states	**COMPILED**	Rules are states in the rule space
No self-containing system can be complete	**COMPILED**	From C1 plus the rule extension
Relation to formal incompleteness theorems	**OPEN**	Different objects, different methods; untested
Gnomon arithmetic inherits float error behaviour	**OPEN**	Correspondence structural only
A constraint selects the radix	**OPEN**	Free choice in both systems examined
The balance condition has a general form	**OPEN**	Stated for two systems only
This account ever completed	**FORBIDDEN**	Completion pins the rule; C1 excludes it

Appendix: Live Output, Quoted Verbatim

Section 3 — the base case, exhaustive.

```
n=2:  1 C1-legal maps,  1 of them bijective  -> C1 ALONE forces the bijection
n=3:  8 C1-legal maps,  2 of them bijective  -> gap of 6
n=4: 81 C1-legal maps,  9 of them bijective  -> gap of 72
```

Sections 5 and 6 — the pinned rule and the floating rule, identical bit count.

```
Q8.8 fixed-point format, 16 bits total
 max representable : 127.99609375  -> 128.0 OVERFLOWS, cannot exist
 min representable : 0.00390625    -> 0.001 UNDERFLOWS to 0, distinction LOST
 dynamic range     : 3.277e+04
float16, also 16 bits
 dynamic range : 1.073e+09
 ratio vs fixed: 3.275e+04x larger
```

Section 7 — channel separation under radix scaling.

```
0.75 x 2^-200: mantissa=0.750000  exponent= -200
0.75 x 2^-100: mantissa=0.750000  exponent= -100
0.75 x 2^  +0: mantissa=0.750000  exponent=   +0
0.75 x 2^+100: mantissa=0.750000  exponent= +100
0.75 x 2^+300: mantissa=0.750000  exponent= +300
exponent spread while sweeping mantissa 0.50 -> 0.99 : 0
```

Section 8 — renormalization frequency.

```
multiplications where the exponent had to be re-normalized: 38691/100000 = 38.7%
```

Section 9 — the denormal region.

```
     value   normal?   ulp (precision)
  1.000e-300    True      1.658e-316
  1.000e-310    False     4.941e-324
  1.000e-320    False     4.941e-324
  4.941e-324    False     4.941e-324
   0.000e+00      --    DISTINCTION GONE
```

Section 10 — the gnomon bound and the two coordinates.

n	shell k	offset m	shell cap 2k	value jump	shell jump
8	2	4	4	5	1
24	4	8	8	9	1
48	6	12	12	13	1

n=1..199999: offset out of range [0,2k] in 0 cases